

# iClusterBayes Summary Report

Automated Report

## Hyperparameters

```
cat('sdev = 0.05, beta.var.scale = 1\n')
```

```
## sdev = 0.05, beta.var.scale = 1
```

## Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.2194  0.2439  0.2517  0.2517  0.2589  0.2889

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09833 0.14111 0.13813 0.17222 0.88278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.1022  0.1450  0.1418  0.1767  0.7239
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.101992 0.031111 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.077249 0.023889 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.006111 0.015000 0.162307 0.118333 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04278 0.05944 0.06333 0.06315 0.06722 0.08722

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.06333 0.10833 0.11277 0.14722 0.89667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.06625 0.11278 0.11623 0.15222 0.67167
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001667 0.060000 0.415376 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0050000 0.2483599 0.3805556 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1294 1.0000 0.6982 1.0000 1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01444 0.02611 0.03056 0.03089 0.03500 0.05833

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02667 0.07056 0.07914 0.11278 0.85222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02778 0.07444 0.08240 0.11778 0.60833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.9189  1.0000  0.7933  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0005556 0.2238627 0.1551389 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05389 1.00000 0.67336 1.00000 1.00000

--- Summary for K = 4 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.002222 0.007778 0.009444 0.010160 0.012222 0.030556

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03056 0.06667 0.07649 0.10389 0.82667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.03333 | 0.07111 | 0.08063 | 0.11111 | 0.68667 |

| Modality 2 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.9361  | 1.0000 | 0.7971 | 1.0000  | 1.0000 |

| Modality 3 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 4 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.0000  | 0.0000 | 0.2543 | 0.5308  | 1.0000 |

| Modality 5 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 1.0000  | 1.0000 | 0.8425 | 1.0000  | 1.0000 |

--- Summary for K = 5 ---

Z.ar:

| Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.0005556 | 0.0050000 | 0.0072222 | 0.0076276 | 0.0094444 | 0.0466667 |

beta.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.02111 | 0.05778 | 0.06776 | 0.09611 | 0.76111 |

| Modality 2 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 3 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 4 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 5 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.02333 | 0.06333 | 0.07259 | 0.10389 | 0.59111 |

| Modality 2 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.6694  | 1.0000 | 0.7711 | 1.0000  | 1.0000 |

| Modality 3 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 4 : |  |  |  |  |  |
|--------------|--|--|--|--|--|
|--------------|--|--|--|--|--|

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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2873  0.9133  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000  1.0000  0.8982  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001111 0.006667 0.008333 0.008843 0.010556 0.027778

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.01278 0.04667 0.05972 0.08556 0.79111
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.01444 0.05111 0.06395 0.09333 0.57056
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.3372  0.4553  1.0000  1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8917 1.0000  1.0000  1.0000  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2276  0.3649  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.9194  1.0000  0.8313  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0005556 0.0044444 0.0061111 0.0067271 0.0083333 0.0250000

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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0.00000 0.01056 0.03778 0.05240 0.07444 0.75944
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.01222 0.04222 0.05722 0.08333 0.52333
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.05000 0.4181  1.0000  1.0000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.204194 0.008194 1.000000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.6533  1.0000  0.7649  1.0000  1.0000

--- Summary for K = 8 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.001111 0.005556 0.007778 0.008400 0.010556 0.034444

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.008889 0.036667 0.053713 0.076111 0.763333
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

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gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01000 0.04222 0.05841 0.08667 0.50833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.1400  0.4153  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.4972  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1802  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3544  0.9178  0.6819  1.0000  1.0000

--- Summary for K = 9 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.003333 0.008333 0.010556 0.010679 0.012778 0.027222

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001667 0.016111 0.034721 0.050000 0.671667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.002222 0.018889 0.038832 0.058333 0.446667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2846  0.6339  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.9865  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.06503 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.

```

```
0.0000 0.0000 0.3939 0.4465 0.9556 1.0000
', '\n`'))
```

```
## ‘‘‘
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2194 0.2439 0.2517 0.2517 0.2589 0.2889
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09833 0.14111 0.13813 0.17222 0.88278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1022 0.1450 0.1418 0.1767 0.7239
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.101992 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.077249 0.023889 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015000 0.162307 0.118333 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.04278 0.05944 0.06333 0.06315 0.06722 0.08722
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.06333 0.10833 0.11277 0.14722 0.89667
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.06625 0.11278 0.11623 0.15222 0.67167
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001667 0.060000 0.415376 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0050000 0.2483599 0.3805556 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1294 1.0000 0.6982 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.01444 0.02611 0.03056 0.03089 0.03500 0.05833
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02667 0.07056 0.07914 0.11278 0.85222
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02778 0.07444 0.08240 0.11778 0.60833
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9189  1.0000  0.7933  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0005556 0.2238627 0.1551389 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05389 1.00000 0.67336 1.00000 1.00000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.002222 0.007778 0.009444 0.010160 0.012222 0.030556
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03056 0.06667 0.07649 0.10389 0.82667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03333 0.07111 0.08063 0.11111 0.68667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9361  1.0000  0.7971  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2543  0.5308  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.8425  1.0000  1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0005556 0.0050000 0.0072222 0.0076276 0.0094444 0.0466667
##
## beta.ar:
##   Modality 1 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.00000 0.02111 0.05778 0.06776 0.09611 0.76111
##   Modality 2 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##   Modality 3 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##   Modality 4 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##   Modality 5 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.00000 0.02333 0.06333 0.07259 0.10389 0.59111
##   Modality 2 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.0000 0.6694 1.0000 0.7711 1.0000 1.0000
##   Modality 3 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##   Modality 4 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.0000 0.0000 0.0000 0.2873 0.9133 1.0000
##   Modality 5 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.0000 1.0000 1.0000 0.8982 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.001111 0.006667 0.008333 0.008843 0.010556 0.027778
##
## beta.ar:
##   Modality 1 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## 0.00000 0.01278 0.04667 0.05972 0.08556 0.79111
##   Modality 2 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##      1      1      1      1      1      1
##   Modality 3 :
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01444 0.05111 0.06395 0.09333 0.57056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.3372  0.4553  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8917  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2276  0.3649  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9194  1.0000  0.8313  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min.  1st Qu.    Median      Mean  3rd Qu.      Max.
## 0.0005556 0.0044444 0.0061111 0.0067271 0.0083333 0.0250000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01056 0.03778 0.05240 0.07444 0.75944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.04222 0.05722 0.08333 0.52333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0500  0.4181  1.0000  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.204194 0.008194 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.6533  1.0000  0.7649  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001111 0.005556 0.007778 0.008400 0.010556 0.034444
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.036667 0.053713 0.076111 0.763333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01000 0.04222 0.05841 0.08667 0.50833
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.1400  0.4153  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4972  1.0000  1.0000  1.0000  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1802  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.3544  0.9178  0.6819  1.0000  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.003333 0.008333 0.010556 0.010679 0.012778 0.027222
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.001667 0.016111 0.034721 0.050000 0.671667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.002222 0.018889 0.038832 0.058333 0.446667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2846 0.6339 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 1.0000 1.0000 0.9865 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06503 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.3939 0.4465 0.9556 1.0000
##
## ‘ ‘ ‘

```

## Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.